Effects of verb type and prior context on the production of relative clause extraposition in English

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Introduction: English speakers have a choice between a complex discontinuous structure, known as relative clause extraposition (RCE), and a simpler adjacent (ADJ) order (1a-b).

1. (a) A reporter showed up who wanted to ask questions. (RCE)
   (b) A reporter who wanted to ask questions showed up. (ADJ)

Why should speakers ever choose (1a)? Early studies observed that RCE is commonly used with intransitive 'appearance' verbs (e.g. *show up*) to express a presentational focus on the subject (Guéron, 1980; Huck & Na, 1992). Rochemont and Culicover (1990, p. 65) further observed that RCE with non-appearance verbs as in (2a) is degraded but becomes more acceptable when the same verb has been recently mentioned and thus is backgrounded (2b).

2. (a) A man screamed who wasn't wearing any clothes.
   (b) Suddenly there was the sound of lions growling. Several women screamed. Then a man screamed who was standing at the very edge of the crowd.

Consistent with these proposals, Francis and Michaelis (2014) found that the rate of RCE increased from 15% to 53% when the main verb or its synonym occurred in the prior context. Similar context effects have been shown experimentally for RCE in German (Bader, 2024; Poschmann & Wagner, 2016), but no prior studies have tested Rochemont and Culicover's claim that context should interact with verb type. In this study, we use a sentence construction task to investigate this interaction. We expect participants will use RCE more often when the verb has recently been mentioned and that this effect will be greater with non-appearance verbs, since appearance verbs are inherently compatible with the presentational function of RCE.

Method: 50 adult L1 American English speakers will be recruited via Prolific. Data collection and coding are ongoing, with preliminary data from ten participants reported here. Each participant produced 32 experimental responses (plus 48 fillers) across four conditions of Verb Type (appearance vs. unergative) (Table 1) and Context Type (given verb vs. new verb). On each trial, participants read aloud a brief context sentence and were then presented with three fragments: a subject NP (always in the middle position), a verb phrase, and a relative clause. The order of the two non-subject fragments was randomized. Participants silently arranged the fragments into a natural continuation of the context and spoke the sentence (Figure 1). Spoken responses were later transcribed and coded for structure choice (RCE vs. ADJ).

Preliminary Results and Discussion: A mixed-effects logistic regression with crossed random intercepts for participant and item revealed a significant main effect of Verb Type ($\beta = -1.70$, $p < .001$), indicating that structure choice varied significantly across verb types. Appearance verbs elicited substantially more RCE structures (27.2%) compared to unergative verbs (4.0%). No main effect of Context Type was shown ($\beta = 0.68$, $p = .090$), nor was any significant interaction between Verb Type and Context Type observed ($\beta = -1.04$, $p = .170$) (Figure 2). These results for verb type are consistent with previous findings from a corpus study (Francis & Michaelis, 2014) and an acceptability judgment task (Walker, 2013). However, the expected effect of context is not shown in this preliminary dataset. The full dataset, once collected, will inform our understanding of how verb meaning and discourse context enhance or diminish the accessibility of RCE during production to enable efficient communication (Levshina, 2022; MacDonald, 2013).

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Unergative verbs	Appearance verbs
jump, shout, cheer, dance, laugh, scream, wave, smile	enter, approach, show up, speed past, emerge, arrive, materialize, appear

Table 1. Verbs used in the sentence construction task

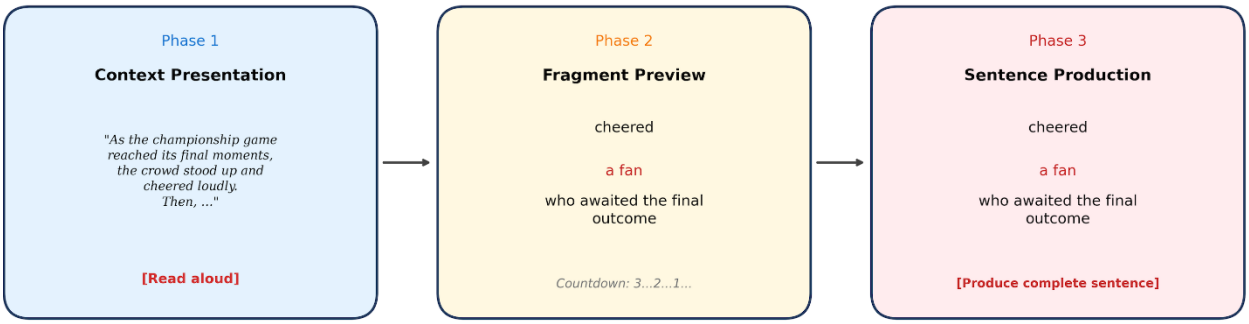

Figure 1. Task flow for the sentence construction task (unergative-given condition shown here)

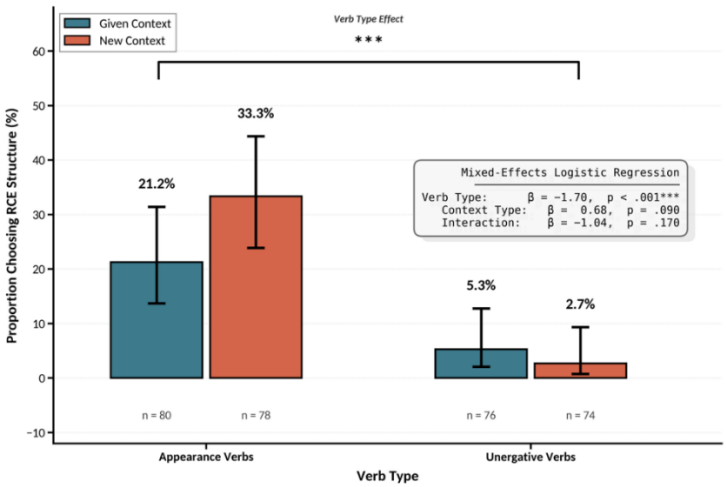

Figure 2. Effect of verb type and context type on structure choice

References

Bader, M. (2024). Relative clause extraposition and information structure. In A. Himmelreich, D. Hole and J. Mursell (Eds.),*To the left, to the right, and much in between: A Festschrift for Katharina Hartmann* (pp. 205-216). Frankfurt: Goethe University Frankfurt.

Francis, E. J., & Michaelis, L. A. (2014). Why move? How weight and discourse factors combine to predict relative clause extraposition in English. In B. MacWhinney, A. Malchukov, & E. Moravcsik (Eds.), *Competing Motivations in Grammar and Usage*. Oxford University Press.

Guéron, J. (1980). On the syntax and semantics of PP extraposition. *Linguistic Inquiry*, *11*(4), 637-678.

Huck, G. J., & Na, Y. (1992). Information and contrast. *Studies in Language*, *16*(2), 325-334.

Levshina, N. (2022). *Communicative Efficiency*. Cambridge University Press.

MacDonald, M. C. (2013). How language production shapes language form and comprehension. *Frontiers in Psychology*, *4*, 40296.

Poschmann, C., & Wagner, M. (2016). Relative clause extraposition and prosody in German. *Natural Language & Linguistic Theory*, *34*, 1021-1066.

Rochemont, M. S., & Culicover, P. W. (1990). *English Focus Constructions and the Theory of Grammar*. Cambridge University Press.

Walker, H. (2013). Constraints on relative clause extraposition in English: An experimental investigation. In *Rightward movement in a Comparative Perspective* (pp. 145-172). John Benjamins Publishing Company.